Portfolio Overview: Strategic UX & Research Operations

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Focus: Data-Driven Product Strategy & Technical Research Infrastructure

Here are some portfolio case studies to showcase my different approaches to user research, focusing on my UX research methodologies and data-driven design decisions.

Case Study 1: Killing the #Tag: Designing "Invisible Organization" for Comedy OS

Why forcing users to organize their own data is a UX failure — and how Comedy OS uses **Semantic Search** and **Progressive Disclosure** to fix it.

Stand-up comedy is an art form of fleeting moments. A comedian might get a brilliant premise for a joke while driving, in the shower, or waiting in line at the store.

I am currently building **Comedy OS** — a performance intelligence platform and digital workspace for comedians. Working as a solo founder, my primary directive for the app has always been simple: **Zero Friction.** If the app takes more than three seconds to open and capture a thought, the joke writing process is in trouble.

But as I mapped out the user flows, I hit a classic trap in productivity software design: **The categorization problem**.

How do I build a system that lets a user instantly dump a raw thought, but still allows them to find that exact thought three years later when they are reworking old jokes that have potential but are not hitting quite yet?

The industry-standard of note taking apps answer is the #tag or the folder. I decided to kill them both. Here is how I navigated the UX and technical architecture with the help of Gemini, my AI co-pilot.

Research & The Problem

The Cognitive Friction of the Filing Cabinet
Imagine a comedian types a quick note:
"The TSA made me take off my shoes, but let a guy with a falcon walk right through."
In a traditional app, the user must now pause and become a librarian.
Should I tag this #airline, #airplane, #airport, or #travel?

This is cognitive friction. As a product designer, I knew this forced the user out of the **creative** *"flow state"* and into an **administrative** *"filing state."* Ultimately, it leads to tag fatigue. The user abandons the process of writing and improving the joke because the coginitive load of organizing is smashing into the flow state of writing. They are governed by different mental systems and need to be separated. While writing the comedians needs to stay in flow state. When the flow state is broken by the app constantly the user abandons the writing and even the app itself.

I wanted a **“Create and Forget”** workflow to minimize the cognitive drain during the writing process. I wanted an **Invisible Librarian**.

Research

I have analyzed over 200 articles from technical writers, hobbyists and journalists on how they use note taking apps for their work and personal projects. While many articles complain about the UI of most apps for certain tasks a large majority mentions the “**organizational block”** that comes with many systems like OneNote and Notion. The user is forced to play the librarian before being able to write. In my own day to day work I have noticed similar things where most apps force me to organize before I can create. Logseq has a journaling approach that is much more helpful to a “**create and forget**” type of workflow. It still uses a sort of “tagging” system with its bi-directional links *[[topic name]]* the user needs to actively use which is not much different from using #*tags*.

In addition I was lucky to talk to my friend Hal who is a stand up veteran of over 6 years. He was able to clarify and validate many ideas I had. Currently he is my main guiding persona for the app. Once Comedy OS is in alpha testing more personas are possibly to come out of the user data.

The Technology: Gemini Suggests Semantic Search

I took this “**organizational block”** UX roadblock to Gemini. I explained that I wanted users to be able to find their notes without ever doing the homework of tagging them.

Gemini suggested abandoning exact-keyword matching entirely and implementing **Semantic Vector Search** using Google Cloud's Vertex AI.

Gemini helped me architect the backend logic: When a comedian saves a note, a Cloud Function silently asks the AI to convert the text into a "vector embedding" — a mathematical representation of the *meaning* of the joke.

The result is magical. Six months later, the comedian can open the search bar and type: *"Find that bit about airport security."* Even though the words "airport" and "security" are nowhere in the original text, the AI understands the semantic relationship. It finds the exact note instantly.

I provided the vision; Gemini provided the invisible infrastructure.

The UX Dilemma: My Pivot on the Tagging Addiction

Technically, the problem was solved. But from a UX perspective, my intuition told me I had a new challenge: **Human habit.**

Users coming from legacy tools like Evernote, Apple Notes, or Notion are conditioned to rely on tags. I realized that if a power user downloads Comedy OS and finds no way to type a `#tag`, they will feel naked and out of control. They might assume the app is too basic for their needs.

I needed a bridge between the old way of working and the new "**Invisible Librarian.**"

The Solution

Progressive Disclosure and The “Intervention”

I brainstormed with Gemini on how to bridge this gap without cluttering the UI. We landed on a UX strategy called **Progressive Disclosure**.

I decided to leave the manual tagging functionality completely intact. If a user wants to type `#heckler` at the bottom of their note, the system happily accepts it.

But here is where Gemini and I combined forces to build an “Intervention.” Gemini helped me write the database logic to add a silent counter to the user’s profile: `manual_tag_count`.

I designed the system to let the user work exactly how they are comfortable. But the moment their `manual_tag_count` hits 10, the system realizes: *This user is doing too much admin work.*

```
+----------------------------+
|           User             |
+----------------------------+
| - userId: String           |
| - settings: Map            |
| - manualTagCount: Int      | <--- NEW: Tracks legacy behavior
| - hasSeenSemanticTutorial: Bool| <--- NEW: Prevents spamming the UI
+----------------------------+
| + createNote()             |
| + getJournal()             |
| + incrementTagCount()      | <--- NEW
+----------------------------+
              |
              | 1
              |
              | *
```

Part of the Class Diagram of the new system

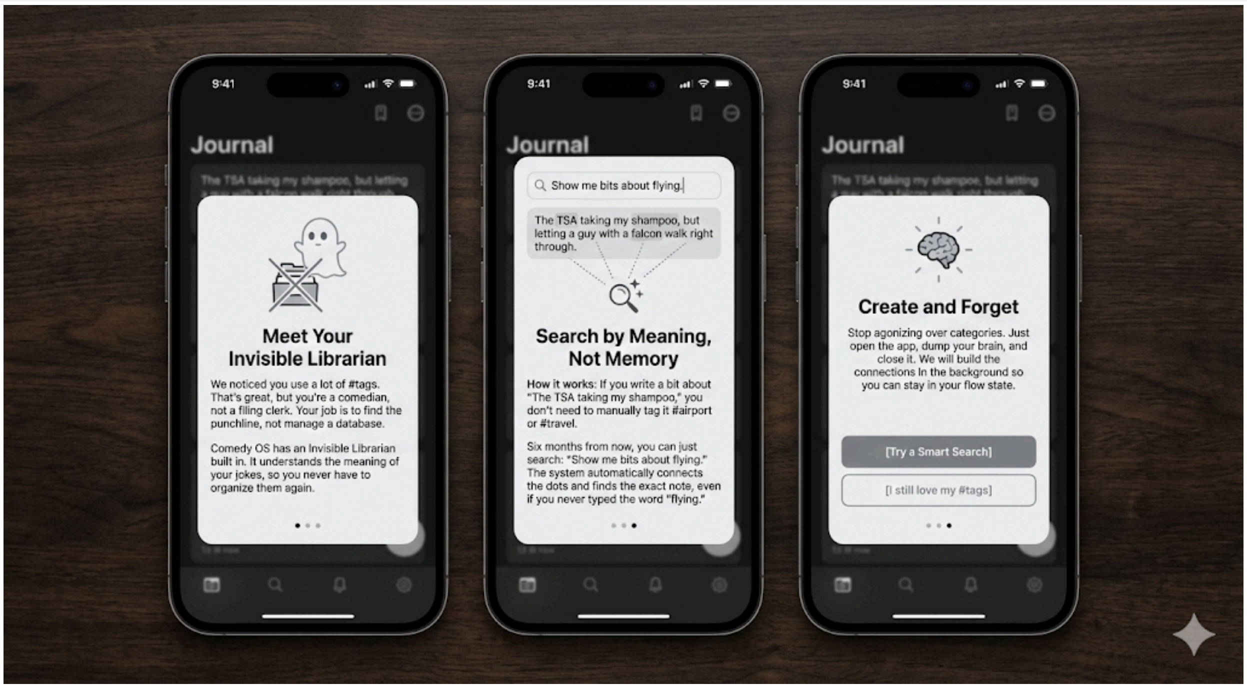
The next time they open their Journal, they aren't hit with an annoying pop-up. Instead, they see an elegantly designed, dismissible card I mapped out for the top of their feed:

You're doing too much admin work.

Button: [See a better way.]

When they click it, a crisp tutorial explains the Invisible Librarian. It shows them that they don't ***need*** to tag things anymore because they can just search by meaning.

Press enter or click to view image in full size



The 3-slide tutorial appears when the system notices the user overusing tags and suggests a better way. It is written to be punchy, highly scannable, and directly address the comedian's desire to avoid administrative "homework."

The Takeaway for Solo Builders

Building a product alone doesn't mean you have to solve every problem alone. By leaning on my design intuition for the *user experience* and relying on Gemini for the *technical architecture*, I was able to build a feature that achieves the best of both worlds:

1. **Zero friction for new ideas:** The "Create and Forget" workflow remains pure.
2. **Respect for legacy habits:** Power users aren't alienated.
3. **Organic education:** Users discover the platform's AI capabilities exactly at the moment they need them most.

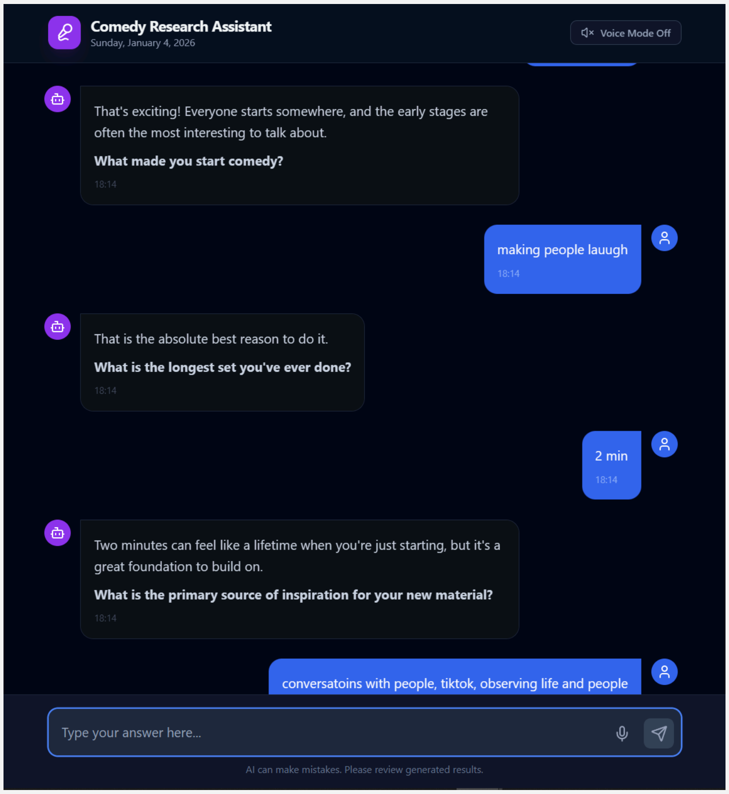
In the end, my job wasn't to build a better filing cabinet. It was to realize that in the age of AI, the filing cabinet is obsolete. Just write the joke. The system will handle the rest.

Case Study 2: Comedy OS Researcher Chatbot

Project Scope: Research Operations (ReOps), Custom Tooling, & Technical Prototyping Core Concept: Scaling Qualitative Research via AI-Driven, GDPR-Compliant Infrastructure.

Overcoming Time Zones with an Asynchronous Research Agent by offloading the interview process to an AI instead of waiting for comedians to get back to me for a scheduled video call.

Project Link: Comedy Research Agent (Google AI Studio)



Solving the Comedy OS Challenge via AI Research

Developing *Comedy OS*—a note-taking app designed to help comedians overcome writer's block by measuring audience reactions and delivering data-driven success metrics—required deep user insights. However, a major logistical barrier emerged: I am based in Central Europe (CET), while the target demographic is in the US (EST/PST). This time zone difference and reliance on unpaid goodwill made synchronous interviews inefficient and prone to ghosting. To resolve this, I created an **Asynchronous Research Agent** with the help of Google Cloud Platform Tools, Google Apps Script and Gemini PRO.

1. The Research Ops Challenge

Traditional surveys (Typeform/Qualtrics) yield shallow data. Human interviews yield deep data but are unscalable and expensive.

- **The Goal:** create a scalable method to conduct "Deep Dive" interviews without the linear constraints of a static survey.

2. Technical Solution: Serverless AI Architecture

To manage this time difference and scale research, I built a custom **Research Agent**. This wasn't a static survey; it was a conversational interface designed to mimic a skilled interviewer. This custom web application uses **React** and the **Gemini API** to conduct autonomous user interviews. It was trained on my previous research to improvise and ask nuanced follow-up questions.

- **Dynamic Inquiry:** Unlike a static form, the AI adapts follow-up questions based on the user's previous answers, probing for "Why" rather than just "What."
- **Live Data Streaming:** To ensure zero data loss, I implemented a "Live Logging" stream. Every user interaction is timestamped and pushed instantly to a private Google Sheet, allowing for real-time monitoring of study progress.
- **System Synchronization:** Solved the challenge of Large Language Models "forgetting" their place in a long survey by storing all questions in a constant frontend array, actively ensuring the UI progress bar and the AI’s flow remain in sync for a smooth, structured interview experience.
- **Key Insight:** Comedians are often night owls who write spontaneously. A conversational agent available 24/7 could capture "in-the-moment" frustrations that a static survey would miss. (The next phase will focus on synthesizing actionable findings from the collected data to drive the app’s feature design.)

3. Compliance & Data Privacy (GDPR Focus)

Designing for the European market requires strict adherence to data privacy.

- **Privacy by Design:** I engineered the system to be **fully anonymous**. The custom React frontend allows public access without login credentials.
- **Risk Mitigation:** By decoupling the interview interface from user identity databases, we minimized PII (Personally Identifiable Information) exposure, ensuring the research

remains compliant with strict ethical and legal standards (GDPR).

4. Impact

- **Scale:** Enabled simultaneous interviewing of unlimited participants, removing the "Researcher Time" bottleneck.
- **Data Richness:** Early testing shows a significant increase in response depth (word count) compared to static text fields, as the conversational UI encourages elaboration.

Current Status

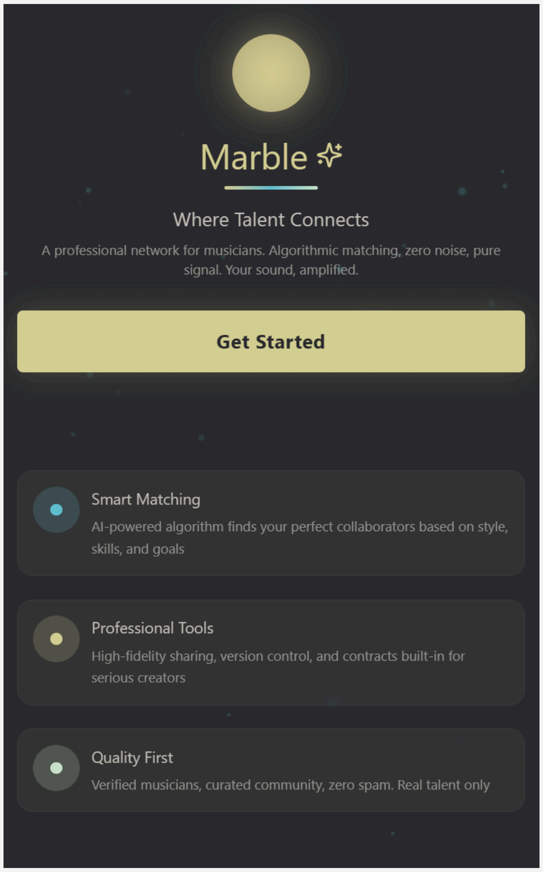
The project is **fully functional**. I now have a live React web application that conducts a professional interview using Gemini, instantly logs every word to a private Google Sheet, requires no user login, and protects my proprietary research data.

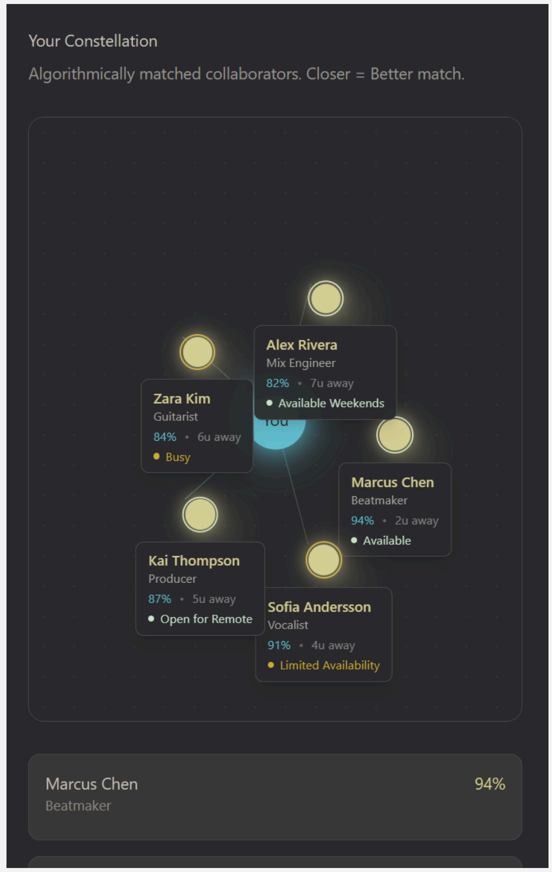
Case Study 3: Marble Music Match

Project Scope: *End-to-End Product Strategy, UX Research, & Ecosystem Design*

Core Concept: *Bridging the "Satisfaction Gap" in Music Collaboration via "Invisible UX".*

Project Link: Marble Testing Prototype (Figma Make)





1. The Enterprise Challenge: "Enshittification" & Churn

The music collaboration market (e.g., Vampr, BandLab) suffers from high churn due to "Enshittification"—the degradation of user experience for monetization.

- **Market Signal:** My analysis of the ecosystem revealed a **"Satisfaction Gap."** Users loved the *idea* of collaboration but hated the *execution* (bots, paywalls, friction).
- **The Business Problem:** Competitors focused on "Social Features" (vanity metrics) while ignoring "Production Workflows" (retention metrics).

2. Data-Driven Methodology (The "Quant-to-Qual" Bridge)

I rejected the standard "User Interview" starting point. Instead, I built a quantitative foundation to ensure our qualitative testing was unbiased.

- **Big Data Audit:** I scraped and analyzed **12,000+ App Store reviews** of competitor products.

- *Insight:* **49.5%** of users were delighted by the *concept*.
- *Failure Point:* **32.2%** left due to technical friction and execution failures.

- **Strategic Decision:** This 32% "Frustrated Segment" became our primary target market.
- **Monitored Testing Process:** In a live interview via Discord and Google Meet I was able to directly observe users' screens and how musicians who are intrigued by the idea of the product were using the UI and what they were stumbling on.
- **Team:** Of 3 founders (including me, CMO). A developer (CTO) and a business specialist (CFO). We minimize friction and ensure stakeholder inclusion through weekly sessions alongside regular technical calls. I conducted workshops on research and testing, integrating stakeholder feedback into our user testing. Rapid iteration allowed for quick process improvement. We use automation to enhance documentation and manage feature creep.
- **Strategic Prioritization:** We used a MOSCOW matrix, informed by prior research, to prioritize initial beta features. Several features, notably B2B functionality, were moved to later phases. This was strategic: we aimed to secure funding by showing immediate B2C feature-market fit (connecting creatives), which user feedback confirmed was essential for platform viability.
- **B2B Considerations:** The app's future direction is B2B, focusing on recording studios, event locations, music labels, and talent agents, though this is not in the current prototype. Planned B2B features include localization, compliance, security, and specialized workflows (e.g., financial planning, logistics). The next phase will test the "Talent Agent" workflow. The current MVP focuses on connecting and enabling collaboration between music creatives.
- **Our Iterative Cycle:** *Hypothesis > Research > Synthesis > Design > Validation > Launch > Post-Launch Research*

  We **collect** our **assumptions** > **find** partial **confirmation** in **research** (user reviews, market research / competitive research) > synthesize **user personas** > **align** the **team** on useful features and **validate** them in **user tests** > after launching a feature we **repeat** the

process until the MVP is ready to be deployed to the market for a closed beta > in closed beta **collected data** will inform **future development** which will have a similar approach.

3. Solution: "Invisible UX" as a Retention Strategy

Enterprise users (Professional Producers) do not want *more* screen time; they want *efficiency*. I designed "Invisible UX" features to solve specific workflow bottlenecks identified in the persona "Producer 1."

- **The Feature:** Intelligent File Naming (e.g., [Artist]_[BPM]_[Date].wav).
- **The Enterprise Logic:** This ensures **Interoperability**. When a file leaves the mobile app and enters a desktop DAW (Digital Audio Workstation), it retains its context without the user typing a single character. This reduces administrative overhead for the user by ~90%.

Features & Reasoning (Mapped to User Needs)

User Need (from Analysis)	Feature Implementation	Design Reasoning
"Zero-Friction Extraction" Professional producers (Persona: Producer 1) need raw materials (samples) for their DAW immediately, without jumping through hoops.	**"Invisible UX" File Naming** Automatic file naming convention: [ArtistName][VideoTitle][Date].wav upon download.	**Reasoning:** By embedding attribution data (metadata) directly into the filename, the user can drag-and-drop files into Ableton/Logic and retain credit info without opening the app again.
Synergy & Verification Curators (Persona: Layla) need to vet "soft skills" (punctuality,	**Synergy Visualization & Debriefs** Visual markers for shared skills and AI-moderated "Chemistry Debriefs"	**Reasoning:** Reduces the risk of "bad dates." The feature creates a structured layer of trust,

vibe) before committing to a call.	after calls.	addressing the market analysis finding that current platforms are full of "catfish and creeps".

4. Projected Business Impact (Success Metrics)

While the application is currently in the prototype validation phase, our competitor benchmarking and qualitative testing forecast a significant impact on **Unit Economics** and **User Retention** compared to market leaders.

- **Churn Mitigation:** Targeting a **15% reduction in churn** by directly addressing the specific friction points found in the 32% "Frustrated User" segment.
- **Efficiency KPI:** Reduced "Time-to-DAW" (Time from discovery to production-ready usage) from **4 minutes** (competitor avg) to **<60 seconds**.
- **LTV Growth:** Higher workflow integration directly correlates to increased Daily Active Usage (DAU) among professional-tier users.